[METRIC]

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13 November 1979
SUPERSEDING
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26 March 1973

MILITARY SPECIFICATION

INSECTICIDE, MALATHION (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two grades of malathion insecticide, chemically known as O,O-dimethyl phosphorodithioate of diethyl mercaptosuccinate.

1.2 Classification. Malathion shall be of the following grades as specified (see 6.2):

Grade A - Premium.
Grade B - General purpose.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters).

FSC 6840

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 162 - Regulations for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act.
49 CFR 100 to 199 - Department of Transportation Hazardous Materials Regulations.

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publications should cite "40 CFR 162" or "49 CFR 100 to 199.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D891 - Specific Gravity of Industrial Aromatic Hydrocarbons and Related Materials.
D1193 - Reagent Water.
D1533 - Water in Insulating Liquids (Karl Fischer Method).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Chemical and physical characteristics. Malathion shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Requirement		Test paragraph
	Grade A	Grade B	
Purity, percent by weight, minimum	91.0	91.0	4.2.4.1
Diethyl fumarate, percent by weight, maximum	1.5	1.5	4.2.4.2
Acidity (as sulfuric acid), percent by weight, maximum	0.6	0.6	4.2.4.3
Water, percent by weight, maximum	0.1	0.1	4.2.4.4
Methyl mercaptan, parts per million, maximum	30.0	----	4.2.4.5
Iron, parts per million, maximum	10.0	10.0	4.2.4.6
Specific gravity at 25°/25°C	1.225±0.010	1.225±0.010	4.2.4.7

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the malathion produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions in a period of no more than 24 hours provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling for examination of packaging shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For malathion tests. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 1 liter shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for malathion tests

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
Over 150	5

4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled shipping container. Sample containers shall be examined for the following defects using an AQL of 1.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Interior coating missing or not as specified
(f) Exterior paint coating missing or not as specified
(g) Marking incorrect, missing, or illegible

4.2.3.2 For malathion tests. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For container leakage test. The sample unit shall be one container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 1.5 percent defective.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

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4.2.4.1 Purity (see 6.4).

(a) Preparation of internal standard solution. Prepare a 1.2 percent weight per volume solution of m-diphenoxybenzene in chloroform. This solution is stable for 4 weeks if it is kept tightly sealed and under refrigeration. Allow the solution to warm to room temperature before use.

(b) Preparation of standard malathion solutions. Weigh accurately about 170, 200, and 230 milligram (mg) quantities of malathion of known purity into separate preweighed 25-milliliter (ml) volumetric flasks. Add by volumetric pipet exactly 5 ml of the internal standard solution prepared in (a) to each flask. Dilute to the volume mark with chloroform. Label the three standard solutions "A", "B", and "C", respectively. Solution B is the working standard solution for gas chromatography; solutions A and C are used to check linearity of the gas chromatograph. These solutions are stable for 4 weeks if they are kept tightly sealed and under refrigeration. Allow the solutions to warm to room temperature before use.

(c) Preparation of gas chromatography column. Weigh 6.25 gram (g) of "SP 2401" or "OV 210" into a 250-ml beaker and dissolve in 125 ml of ethyl acetate. Stir the solution to obtain a vortex and add 25 g of the solid support material with continued agitation. Filter the slurry through a Buchner funnel using a No. 1 Whatman paper and a gentle vacuum to minimize solvent loss through evaporation. Continue filtration until the drop rate is approximately one per second. Transfer the packing to a "HI-EFF Fluidizer" or equivalent apparatus and dry the packing with a low flow of heated nitrogen by connecting a source of nitrogen through a pressure reducer to the base of the fluidizer and by placing the fluidizer on a controlled temperature hot plate set at 75°C. Continue the drying procedure until solvent vapors can no longer be detected. Take care that the packing is not blown out of the top of the fluidizer.

(d) Column conditioning. Condition the column for 15 hours at 255°C or approximately 20°C below the maximum temperature recommended for the liquid phase. This step should be conducted with the exit end of the column unconnected to the detector but with the carrier gas flowing at the recommended rate. Connect the exit end of the column to the detector and set the control to provide the conditions listed in table III. Allow the instrument to come to equilibrium and then inject 3-microliter aliquots of the solution C prepared in (b) into the chromatograph until constant response is obtained (at least three consecutive injections for which the response ratios agree to within 2 percent).

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TABLE III. Chromatographic conditions (see 6.4)

Characteristic	Condition
Range	1000
Column dimensions	122 cm by 4 mm ID (6 mm OD)
Column material	Borosilicate glass
Column support	100 to 120 mesh "Gas Chrom Q" or "Supelcoport"
Column coating	5 percent "SP 2401" or "OV 210"
Oven temperature	180°C
Injection port temperature	200°C
Flame detector temperature	300°C
Gas flow rates, ml per minute:	
Helium (carrier gas)	30
Hydrogen	35
Air	425
Integrator settings:	
Noise suppression	Maximum
Slope sensitivity up	0.1
Slope sensitivity down	0.1
BL reset delay	0.15
Area threshold	1000
Retention times, minutes:	
Malathion peak	10
Internal standard peak	6
Minimum time between malathion and internal standard peaks	3.5

NOTE: Integrator settings should be adjusted so that the deflections on the slope meter do not exceed ± 50 percent before injection.

(e) Gas chromatograph. Use a gas chromatograph having a high sensitivity flame ionization detector and an electrometer having a sensitivity of at least 10^{-11} amperes driving a 1 millivolt recorder and having a drift of less than 1 percent per hour. It must have an on-packing injection system and an all-glass column. An electronic digital integrator or computer-based area measurements must also be used. The integrator should have independent controls for the selection of up and down slope sensitivities so that start and stop integration points can be selected.

(f) Chromatographic conditions. Recommended conditions for a Hewlett-Packard Model 7600 gas chromatograph are shown in table III. Other equivalent instrumentation may be used but may require modification of conditions in order to obtain good peak shape, adequate resolution, and appropriate retention times.

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(g) Linearity check. Check the gas chromatograph for linearity at least weekly, whenever new standard solutions are prepared, and whenever the column, new or used, is newly installed in the instrument. Employing digital integration for peak area measurements, determine the appropriate setting and injection aliquot (between 2 and 4 microliters) of standard solution B to yield an area count of at least 100,000 counts (optimum electrometer output with acceptable noise level). Use the conditions so determined for all specimens and standards in the set. Inject triplicate aliquots of the determined appropriate volume of standard solutions A, B, and C into the chromatograph, determine the response ratio for each injection, and average the resulting ratios for each solution. Divide the average response ratio for each solution by the corresponding malathion content in milligrams and compare the resulting response factors. These factors should agree to within 2 percent. Factors not within 2 percent indicates either a weighing error in the preparation of one of the standard solutions or instrumental difficulties which must be corrected before proceeding with an analysis of the specimen.

(h) Procedure. Weigh accurately about 200 mg of the specimen into a preweighed 25-ml volumetric flask. Add by pipet exactly 5 ml of m-diphenoxybenzene to each of the volumetric flasks in the set. Dilute to the volume mark with chloroform and mix well. Inject duplicate aliquots of the appropriate volume determined in (g) of standard solution B. The response ratios should agree to within 2 percent. If the precision limit is not met, inject two more aliquots of standard solution B. Failure to meet the precision requirement with the second pair of injections indicates instrumental difficulties which must be resolved before proceeding with the analyses. Inject duplicate aliquots (of the same volume as used in the preceeding step) of each of the specimen solutions. The precision considerations discussed in the preceeding step also apply here. Average the duplicate ratios for each specimen solution. After every two specimen solutions, inject duplicate aliquots of standard solution B. Average the response ratios of the standards immediately before and after the specimen solution. Use this average to calculate the purity of the two specimen solutions. NOTES: The flame may be extinguished by the passage of the solvent peak through the detector; it should be reignited within 30 seconds of injection to allow re-equilibration before the first peak of interest arrives at the detector. With most specimens, sufficient clarification of the solution is obtained by allowing the solution to stand about 15 minutes. Occasionally, specimens will require centrifuging. In either case, a layer of solids will be found floating on the solvent. If a slight positive pressure is applied (with a bulb) to the pipet while it is carefully inserted into the solution, no particles should be removed with the aliquot taken. The inclusion of fine particles in the solution used for injection can result in clogging of the syringe needle.

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(i) Calculations. For each injection calculate the response ratio as follows:

$$\text{Response ratio} = \frac{\text{Area of malathion peak}}{\text{Area of internal standard peak}}$$

Calculate the percent by weight purity as follows:

$$\text{Percent purity} = \frac{ABC}{DE}$$

where: A = Average response ratio for specimen solution,
B = Weight of malathion standard taken in grams,
C = Percent purity of malathion standard,
D = Average response ratio for standard solution B, and
E = Weight of specimen taken in grams.

4.2.4.2 Diethyl fumarate.

(a) Preparation of standard curve. Weigh to the nearest 0.1 mg approximately 7.5 to 7.7 g of highly purified malathion (see 6.5) into a 200-ml volumetric flask and make up to volume with S.D.A Formula No. 2B ethyl alcohol at 25°C (solution A). Weigh 0.150 to 0.155 g of Eastman Kodak white label grade diethyl fumarate to the nearest 0.1 mg, into a 200-ml volumetric flask and make up to volume with 2B ethyl alcohol at 25°C (solution B). Prepare four standard diethyl fumarate solutions as follows: Pipet 25 ml of solution A into each of four 100-ml volumetric flasks and number them 1, 2, 3, and 4. Pipet 5 ml of solution B into flask 2, 15 ml into flask 3, and 25 ml into flask 4. Make each flask up to volume with 2B ethyl alcohol. Each flask contains approximately 1.0 g malathion; flask 1 contains 0.0 percent diethyl fumarate, flask 2 contains approximately 0.4 percent diethyl fumarate, flask 3 contains approximately 1.2 percent diethyl fumarate, and flask 4 contains approximately 2.0 percent diethyl fumarate. Pipet a 15-ml aliquot from each of the four standards into a correspondingly numbered 50-ml volumetric flask; add 5 ml of 3N hydrochloric acid to each; and make up to volume with water at 25°C. Using a Leeds and Northrup Model E Electrochemograph or equal and a dropping mercury electrode with polarographic cells, pour a portion of the acidified solution of flask 1 into the polarograph cell. Pass nitrogen, saturated by passing the nitrogen through a reservoir containing a portion of the remaining solution from flask 1, through the cell for 10 minutes to expel all oxygen. Then, polarize from -0.1 to -1.1 volts versus the saturated calomel electrode. Record the current in microamps. Repeat the procedure for flasks 2, 3, and 4. Plot the diethyl fumarate concentrations (grams per milliliter) versus the corresponding current in microamps. Use this curve for determining the diethyl fumarate content of the malathion samples.

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(b) Procedure. Weigh 0.9 to 1.0 g of specimen, to the nearest 0.1 mg, into a 100-ml volumetric flask and make up to volume with 2B ethyl alcohol at 25°C. Pipet a 15-ml aliquot into a 50-ml volumetric flask, add 5 ml of 3N hydrochloric acid, and make up to volume with water at 25°C. Pour a portion of the acidified solution into a polarographic cell and treat in the same manner as the acidified solutions in (a). Record the current in microamps. Calculate the percent diethyl fumarate as follows:

$$\text{Percent diethyl fumarate} = \frac{C\,(10^5)}{3W}$$

where: C = Concentration in grams of diethyl fumarate per milliliter, for corresponding current in microamps, of specimen, and
W = Weight of specimen in grams.

4.2.4.3 Acidity. Weigh 10 g of the specimen to the nearest 0.01 g and dissolve in acetone using gentle warming if necessary. Add 75 ml of water and titrate immediately with 0.02N sodium hydroxide solution using methyl red as an indicator. Calculate the percent acidity (as sulfuric acid) as follows:

$$\text{Percent acidity} = \frac{4.9AB}{W}$$

where: A = Milliliters of sodium hydroxide solution used,
B = Normality of sodium hydroxide solution, and
W = Weight of specimen in grams.

4.2.4.4 Water content. Determine percent by weight water content in accordance with ASTM D1533.

4.2.4.5 Methyl mercaptan.

(a) Preparation of standard methyl mercaptan solution. Use a 250-ml gas sampling tube with stopcocks at both ends. Set up a gas sampling apparatus by using Tygon tubing to connect one end of the gas sampling tube to a 250-ml gas leveling bulb filled with redistilled mercury. Fasten the tubing securely with copper wire to protect against the strain caused by the weight of the mercury. Use Tygon tubing to connect the other end of the gas sampling tube to a 3-way stopcock and also to connect the 3-way stopcock to a small cylinder of methyl mercaptan. Open both stopcocks of the gas sampling tube and open the 3-way stopcock to the atmosphere. Raise the gas leveling bulb to displace the air from the gas sampling tube and 3-way stopcock. Open the 3-way stopcock to the gas cylinder. Open the valve on the gas cylinder carefully and lower the gas leveling bulb until

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the gas sampling tube is partially filled with gas (this step fills the tubing between the cylinder and the stopcock with methyl mercaptan). Open the stopcock again to the atmosphere and raise the mercury level to the stopcock to flush the methyl mercaptan and air out of the system. Repeat this step in order to remove all the air from the system. Open the 3-way stopcock to the cylinder and fill the gas sampling tube with methyl mercaptan. Maintain the mercury level in the gas leveling bulb at a higher level than that in the gas sampling tube to insure positive pressure. Close the upper stopcock in the gas sampling tube, remove the tubing, and place a septum over the tube exit. Displace the air between the stopcock and the septum by inserting a gas syringe, without a plunger, into this space and then opening the stopcock. The positive methyl mercaptan pressure displaces the small pocket of air instantaneously. The methyl mercaptan can now be sampled through the septum with a gas-tight syringe. Pipet 12.5 ml of 99.5+ percent 2,2,4-trimethylpentane into a 10-ml serum bottle (actual volume of 14.5 ml to cap), stopper with a silicone-coated rubber stopper and seal with an open-centered aluminum seal using a seal crimper. Insert a 10-ml gas syringe into the gas sampling tube, rinse several times, and then draw more than 7.5 ml of methyl mercaptan into the syringe. Withdraw the syringe from the gas sampling tube and adjust the volume to 7.5 ml. Immediately inject the methyl mercaptan into the serum bottle containing the 2,2,4-trimethylpentane. Insert the tip of the needle well below the surface of the liquid so that the methyl mercaptan will dissolve as it bubbles up through the liquid. Withdraw the syringe and vigorously shake the serum bottle for 1 minute. Note the barometric pressure and temperature and calculate the concentration of the standard solution as follows:

$$\text{Concentration of methyl mercaptan in micrograms per microliter} = \frac{267{,}000}{P(273 + T)}$$

where: P = Atmospheric pressure in millimeters of mercury and
T = Temperature in °C.

(b) Preparation of calibration curve. Repeat the procedure in (a) transferring 5.0, 2.5, 1.0, 0.75, 0.50 and 0.25 ml quantities of methyl mercaptan into separate sealed 10-ml serum bottles containing 12.5 ml of 2,2,4-trimethylpentane. Calculate the concentration of these standard solutions as follows:

$$\text{Concentration of solution in micrograms per microliter} = \frac{267{,}000\ A}{7.5P(273 + T)}$$

where: A = Milliliters of methyl mercaptan injected into solution,
P = Atmospheric pressure in millimeters of mercury, and
T = Temperature in °C.

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Using the conditions specified in (c), measure the chromatographic response of a 1-microliter aliquot of each standard solution. Response should be at least 50 percent of full scale deflection. This is obtained by proper adjustment of the attenuation. Plot the response (peak height) against the concentration (micrograms of methyl mercaptan). Prepare a new calibration curve each day that the analyses are performed. The standard methyl mercaptan solutions should be refrigerated when not in use. Standard solutions may be used for a period of 2 weeks after preparation provided they are refrigerated and no leaks have developed in the system to disturb the liquid-vapor equilibrium in the stoppered and sealed serum bottle.

(c) Chromatographic conditions. Use a flame ionization gas chromatograph. Recommended chromatographic conditions for an F & M Model 1609 chromatograph are shown in table IV. These conditions may be modified, as necessary, if other chromatographs are used.

TABLE IV. Chromatographic conditions

Characteristic	Condition
Column dimensions	1.83 meters by 6.35 mm ID
Column material	Stainless steel
Column support	60 to 80 mesh "Gas Chrom Z"
Column coating	10 percent "XE-60"
Injection port temperature	100°C
Column temperature	30°C
Detector temperature	160°C
Gas flow rates, ml per min:	
Helium	120
Hydrogen	60
Air	250
Range	10
Attenuation	1X
Retention time of methyl mercaptan peak	34 to 36 seconds
Record chart speed	38 mm per minute (high speed No. 1)

NOTES: (1) The range and attenuation are equivalent to 4 x 10^{-11} amperes for full scale deflection.

(2) Convert the methyl mercaptan peak height observed to height at attenuation 1 by multiplying peak height by attenuation.

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(d) Procedure. Condition the column before using by injecting a 1.0-microliter aliquot of the specimen into the gas chromatograph at 160°C (temperature of column and injection port) and temperature programming the chromatograph at 5°C per minute until the specimen is completely eluted. Repeat this operation from three to five times. Pipet 15 ml of specimen in a 27-ml serum bottle (actual volume of 32 ml to cap), stopper with a silicone-coated rubber stopper, and seal with an open-centered aluminum seal using a seal crimper. Place in an oven at 56°C for 2 hours. Using a 1-ml gas syringe, inject 0.5 ml of the vapors directly into the chromatograph and measure the chromatographic response using attenuation 1X. Calculate the parts per million (ppm) of methyl mercaptan as follows:

$$\text{Methyl mercaptan, ppm} = 63.3A$$

where: A = Micrograms of methyl mercaptan in the 0.5-ml aliquot of specimen vapors as determined from the calibration curve.

4.2.4.6 Iron.

(a) Standard iron solution. Dissolve 0.100 g of pure iron wire in 10 ml of 10-percent sulfuric acid and 3 ml of nitric acid. Cool, transfer to a 1-liter volumetric flask, and dilute to volume with water. Transfer a 10-ml aliquot to a 100-ml volumetric flask and dilute to volume. One ml of this standard iron solution contains 0.01 mg of iron.

(b) Standard graph preparation. By means of a buret, measure 0.5, 1.0, 2.0, 7.0, 10.0, 15.0, and 20.0 ml portions, respectively, of the standard iron solution into seven 100-ml volumetric flasks. Add to each 10 ml of 1 molar hydroxylamine hydrochloride solution. Heat to just boiling, cool, add a small piece of congo red paper, and add sufficient ammonium hydroxide solution to change the color of the congo red paper to a bluish-red. Add 10 ml of 0.1-percent 1,10-phenanthroline solution. Allow the color to develop for 10 minutes, dilute to volume with water, and mix. Place a blank of the reagents in a suitable spectrophotometer having a 525-nanometer filter and a 2-centimeter cell path. Adjust the instrument to zero. Record the dial readings for each of the above standards and plot them against the concentration in milligrams per 100 ml. Use the resulting graph for all subsequent iron determinations.

(c) Procedure. Weigh 2.0 ± 0.01 g of the specimen into a 100-ml silica evaporating dish which has previously been washed with hydrochloric acid. Using a hood, gently ignite using a small flame and avoiding application of excessive heat to the bottom of the dish. When the specimen ceases burning, heat gently with a meeker burner until easily combustible matter is destroyed. Transfer to a muffle furnace at 800°C. When the fumes cease, close the furnace door and leave the

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dish in the furnace 10 to 15 minutes to destroy the remaining carbon. Remove the dish and cool in a desiccator. Add 3 ml of hydrochloric acid and 3 ml of water. Cover with a clean watch glass and heat 30 minutes on a steam bath. Cool to room temperature and filter through double acid washed dense texture filter paper. (If the filtrate is not clear, refilter using double paper if necessary to make the filtrate clear.) Rinse the filter several times with small portions of water. Add 10 ml of 1 molar hydroxylamine hydrochloride solution and proceed as in the preparation of the graph in (b). Zero the instrument with a reagent blank. Record the dial reading. From the graph prepared in (b), determine the milligrams of iron in 100 ml for the specimen.

$$\text{Iron, ppm} = \frac{1000\ A}{W}$$

where: A = Milligrams of iron and
W = Weight of specimen in grams.

4.2.4.7 Specific gravity. Determine the specific gravity in accordance with ASTM D891.

4.2.5 Container leakage test. Place the sample container in each of the following positions and leave it in each position for a period of 15 minutes:

(a) Upright
(b) Upside down
(c) On one side (or one quadrant)
(d) On one end (or second quadrant)
(e) On other side (or fourth quadrant)

Examine the container after each period for any evidence of leakage.

5. PACKAGING

5.1 Packing. Malathion shall be packed level A or commercial as specified (see 6.2).

5.1.1 Level A. A quantity of 208 (+2 or -0) liters of malathion shall be packed in a drum conforming to type I of PPP-D-729. The entire inner surface of the drum shall be coated with two coats of high baked-on resin which shall neither affect nor be affected by the malathion (see 6.6). The 19-mm plug shall be closed to a torque of 20.3 to 23.0 newton meters. The 51-mm plug shall be closed to a torque of 40.7 to 44.7 newton meters. Gaskets shall be formed from material which shall neither affect nor be affected by the malathion. Each closure shall be furnished with a cap seal. There shall be no evidence of leakage of malathion from the drum when tested as specified in 4.2.5.

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5.1.2 Commercial. A quantity of malathion as specified (see 6.2) shall be packed in accordance with MIL-STD-1188.

5.3 Marking. Level A packs shall be marked in accordance with MIL-STD-129. Commercial packs shall be marked in accordance with MIL-STD-1188. All packs shall be marked in accordance with 49 CFR 100 to 199. All containers shall be marked to show the date of manufacture of the malathion and the lot or batch number.

5.3.1 Labeling. Unit, intermediate, and shipping containers, if used, shall be labeled in compliance with 40 CFR 162 except that directions for use shall be printed directly on the front of the label. Labels shall be approved and registered in accordance with 40 CFR 162. As a minimum, each container shall be labeled with the following information:

(a) The name and percent by weight of each active ingredient
(b) The name and percent by weight of each inactive ingredient
(c) National stock number, military specification number, contract number and lot number
(d) Manufacturer's name and address
(e) EPA registration and establishment number
(f) Directions for use against adult mosquitoes using thermal aerosols or fogs applied by ground equipment and ultra-low volume (ULV) aerial application (fixed wing aircraft and helicopter); adult mosquitoes and stable flies using ULV nonthermal aerosols applied by ground equipment; forest insects using ULV aerial application; and agricultural uses (including adult mosquito and fly control on alfalfa, clover, pasture and range grass, grass and grass hay, and grain crops) and nonagricultural land (wasteland) using ULV nonthermal aerosols applied by aircraft or ground equipment.
(g) Precautionary information:

FOR MILITARY USE ONLY
NOT FOR RESALE UNDER THIS LABEL

CAUTION: KEEP OUT OF REACH OF CHILDREN
Harmful by swallowing, inhalation, or skin contact. Avoid breathing spray mist. Avoid contact with skin. Wash thoroughly after handling. Change contaminated clothing. Do not contaminate food or feed products.

6. NOTES

6.1 Intended use. The 91-percent malathion formulation is intended for use against adult mosquitoes, flies, forest insects and agricultural crop insects using thermal aerosols or fogs applied by ground equipment or ULV nonthermal aerosols applied by aircraft or ground equipment. Grade A malathion is intended for use where it is desirable to minimize odor. Grade B malathion is intended for use in applications where the presence of odor is not an important consideration.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Grade of malathion required (see 1.2).
(c) Level of packing required (see 5.1).
(d) Unit quantity required for commercial packaging (see 5.1.2).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Purity test materials and equipment. Malathion of known purity may be obtained from American Cyanamid Co., Agricultural Research Division, P.O. Box 400, Princeton, NJ 08540. The prepared column packing may be obtained from Supelco, Inc., Supelco Park, Bellefonte, PA 16823 (specify "Pesticide Grade"); Alltech Associates, 202 Campus Drive, Arlington Heights, IL 60004; or Applied Science Laboratories, P.O. Box 440, State College, PA 16801. "SP 2401" is available under catalog No. 02-1282 from Supelco, Inc. A "HI-EFF Fluidizer" is available from Applied Science Laboratories, Inc.

6.5 Highly purified malathion. Highly purified malathion (diethyl fumarate-free) is available from the American Cyanamid Company, Agricultural Research Division, P.O. Box 400, Princeton, New Jersey 08540.

6.6 Container coatings. Lining systems that have been found satisfactory for use with malathion consist of two coats of high-baked, pigmented phenolic resin with a total film thickness of 0.03175 millimeters. Drum manufacturers should be consulted regarding the type of lining they apply.

6.7 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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